CliniScan Lung-Abnormality Detection on Chest X-rays using AI

Enjam Supriya

Table of Contents:

1.Introduction	03
2.Hyperparameter Tuning	04-05
3.Data Augmentation	06
4.Transfer Learning	06
5.Evaluation Metrics	07
6. Grad-CAM Visualization	08
7. Object Detection	09-10
8. Error Analysis	11
9.Conclusion	11

1.Introduction:

This project focuses on detecting and classifying chest diseases using deep learning techniques. The objective is to improve model performance through hyperparameter tuning, data augmentation, and transfer learning. Additionally, model interpretability is analyzed using Grad-CAM, and object detection is performed using YOLO.

2.Hyperparameter Tuning:

Hyperparameter tuning is a crucial step in improving the performance of deep learning models. It involves adjusting various training parameters to achieve better accuracy, stability, and generalization.

In this project, multiple experiments were conducted by modifying key hyperparameters of the ResNet50 model. The goal was to identify the best combination of parameters that improves classification performance on medical images.

2.1 Parameters Tuned

The following hyperparameters were selected for tuning:

Learning Rate: Controls how quickly the model updates its weights during training.

Batch Size: Determines the number of samples processed before updating the model.

Optimizer: Defines the method used to update model weights (Adam, AdamW).

Number of Epochs: Specifies how many times the model sees the entire dataset.

2.2 Experiment Results

Experiment	LR	Batch	Optimizer	Epochs	Accuracy	Precision	Recall	F1 Score
Baseline	0.001	16	Adam	20	0.748	0.360	0.368	0.362
Experiment1	0.005	32	Adam	20	0.758	0.360	0.368	0.362
Experiment2	0.005	16	Adam	30	0.757	0.379	0.374	0.363
Experiment3	0.005	32	AdamW	20	0.7514	0.3861	0.3929	0.3825

2.3 Analysis

The model performance improved slightly after hyperparameter tuning.

Lower learning rate and increased batch size helped improve training stability.

AdamW optimizer provided better generalization compared to Adam.

3. Data Augmentation

To improve model generalization, data augmentation techniques were applied:

Horizontal Flip

Rotation (±10°)

Brightness and Contrast Adjustment

CLAHE

Noise Injection

These techniques helped the model learn robust features.

4. Transfer Learning

Transfer learning was applied using a pretrained ResNet50 model.

Early layers were frozen

Later layers were fine-tuned

Final classification layer was replaced

This improved feature extraction and reduced training time.

5.Evaluation Metrics

The model was evaluated using:

Accuracy

Precision

Recall

F1 Score

6.1 Results

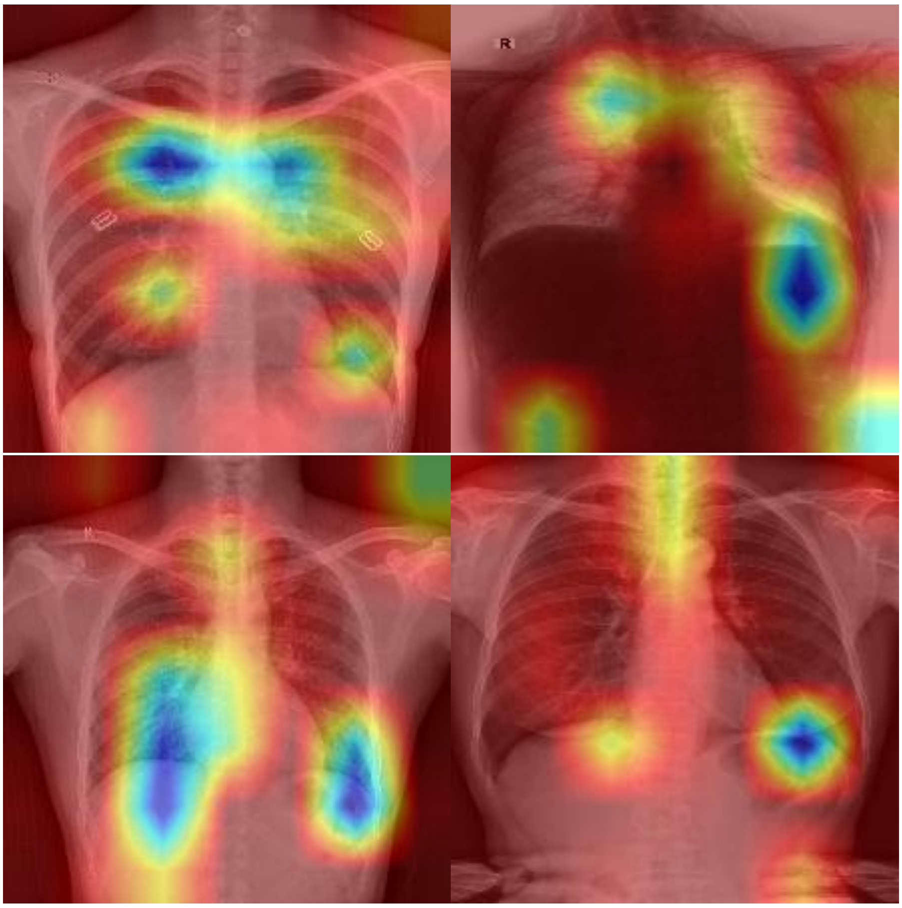
Metric	Value
Accuracy	0.7514
Precision	0.368
Recall	0.3929
F1 Score	0.3825

6. Grad-CAM Visualization

Grad-CAM was used to visualize the regions influencing model predictions.

Images:

Correct predictions



Explanation

Grad-CAM highlights important regions in the image that contribute to the model's decision.

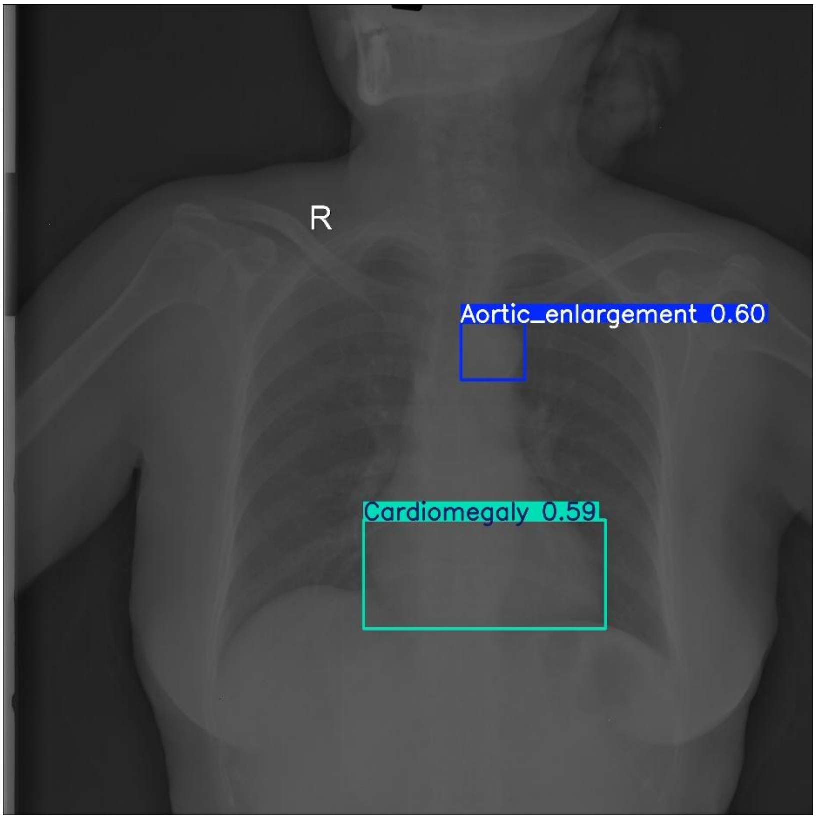
It helps in understanding whether the model focuses on relevant areas.

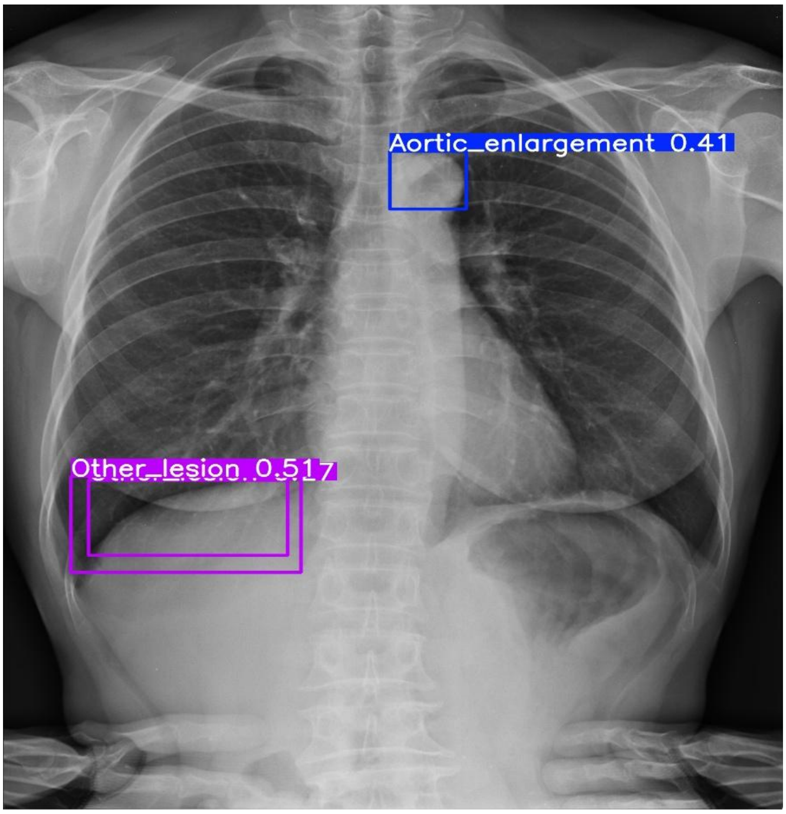
7. Object Detection (YOLO)

YOLO model was used for detecting disease regions in images.

Add Images:

Detection outputs





Explanation

YOLO identifies regions of interest using bounding boxes and confidence scores.

8. Error Analysis

The following observations were made:

Model struggles with low contrast images.

Small abnormalities are sometimes missed.

Some false positives occur in normal images.

Detection bounding boxes are not always precise.

9. Conclusion

The model performance improved through hyperparameter tuning and data augmentation.

Grad-CAM provided interpretability, and YOLO enabled detection of disease regions.

Further improvements can be achieved with more data and advanced models.